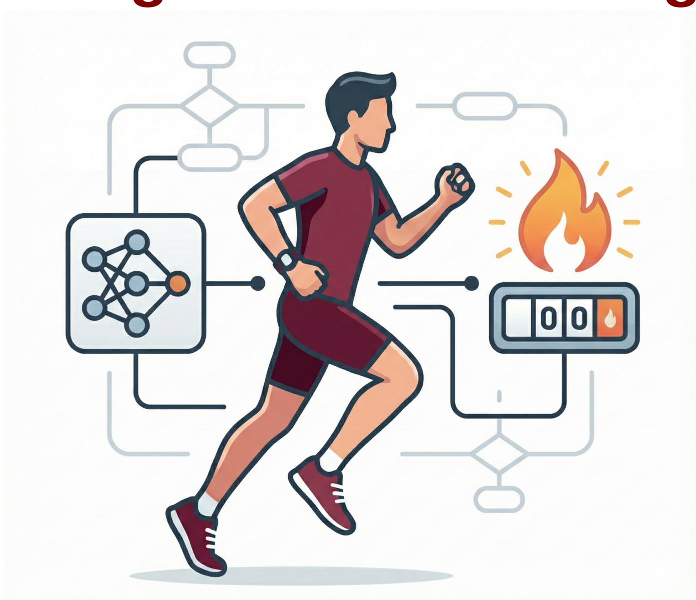
Thompson Rivers University
COMP 4980: Machine Learning

Predicting Workout Calories Burned Using Machine Learning



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Project Overview: This report details the development of machine learning models to predict calorie expenditure during exercise. Utilizing a dataset of 20,000 records, we explore feature engineering, perform extensive data analysis, and evaluate multiple regression models. Our results highlight Gradient Boosting as the optimal solution ($R^2 = 0.997$), demonstrating that sequential error correction significantly outperforms baseline ensemble methods.

1 Data Description

1.1 Dataset Overview

For this project, we used a fitness and lifestyle dataset obtained from Kaggle [1]. The dataset contains information about workout sessions, including physiological measurements, exercise details, and nutritional data.

Dataset Link: Predicting Workout Calories Burned - XGBoost

```
Dataset Characteristics:
  • Size: 20,000 records
  • Features: 54 columns (after initial processing)
  • Target Variable: Calories_Burned (continuous value)
  • File Size: Approximately 10 MB
```

1.2 Feature Categories

The dataset contains several types of features:

- **Demographic Features:** Age, Gender, Weight (kg), Height (m).
- **Physiological Measurements:**
  - `Max_BPM`: Maximum heart rate during workout
  - `Avg_BPM`: Average heart rate during workout
  - `Resting_BPM`: Resting heart rate
  - `Fat_Percentage`: Body fat percentage
  - `BMI`: Body Mass Index
- **Workout Information:**

- Session_Duration (hours): Length of workout session
- Workout_Type: Type of exercise (HIIT, Cardio, Strength, Yoga)
- Workout_Frequency (days/week): How often person exercises
- Experience_Level: Fitness experience level (1-3)

- **Nutritional Data:** Carbs, Proteins, Fats, Water Intake, Diet Type.

1.3 Data Quality and Cleaning Strategy

We performed a rigorous quality check before analysis to ensure model integrity.

- **Missing Values & Duplicates:** We confirmed the dataset had no missing values and no duplicate records, allowing us to proceed without imputation.
- **Outlier Detection (IQR Method):** We utilized the Interquartile Range (IQR) method as it does not assume a normal distribution. We identified 507 outliers (2.54%).
- **Reasoning for Retention:** We elected *not* to drop these outliers. In the context of fitness, extreme values often represent valid high-intensity long-duration sessions (e.g., marathon training). Removing them would bias the model toward "average" workouts and reduce its ability to predict high-performance scenarios.

2 Data Analysis

2.1 Target Variable Distribution

The target variable (Calories_Burned) has the following statistics:

- **Mean:** 1,280 calories
- **Standard Deviation:** 502 calories
- **Range:** 323 – 2,891 calories

The distribution is roughly normal, which is favorable for regression analysis.

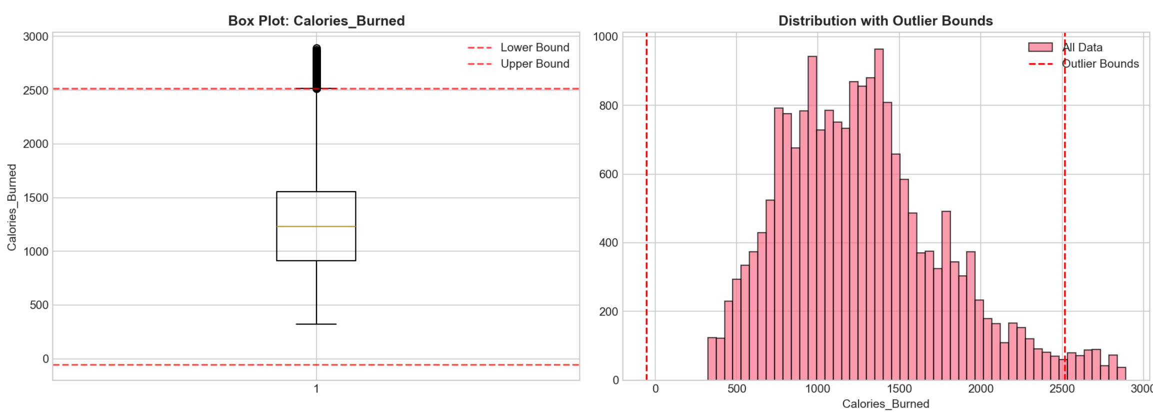

Figure 1: Box plot showing calorie distribution with outlier bounds.

2.2 Correlation Analysis

We calculated correlations between all numeric features and our target variable.

Feature	Correlation with Calories_Burned
Session_Duration (hours)	0.81
Experience_Level	0.70
Workout_Frequency	0.58
Water_Intake	0.26

Table 1: Feature Correlations

Analysis: Session duration showed the strongest correlation (0.81). This validated our data quality, as it aligns with the physiological reality that total energy expenditure is directly proportional to time spent exercising.

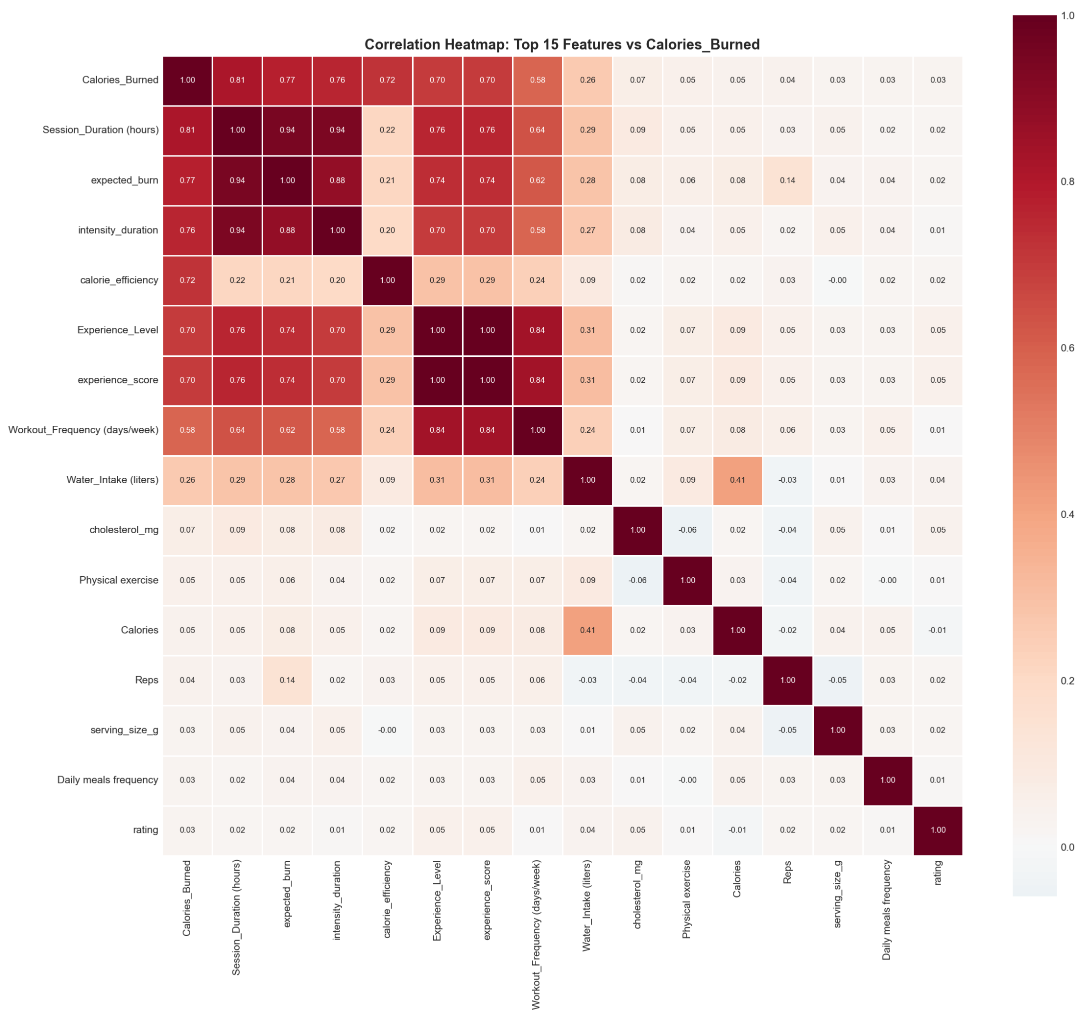

Figure 2: Heatmap showing correlations between top features.

2.3 Feature Engineering

We created specific features to capture physiological interactions that single variables might miss:

1. `intensity_duration` **(Avg_BPM $\times$ Duration):**

- **Reasoning:** Calorie burn is a function of effort (BPM) over time. Multiplying these captures the combined effect.
- **Validation:** This feature achieved a **0.76 correlation** with the target, confirming our hypothesis.

2. `bpm_intensity` **(Avg / Max BPM):**
   - **Reasoning:** intended to represent relative effort percentage.
   - **Result:** Low correlation (0.006). We retained it for completeness, but it suggests absolute heart rate is more predictive than relative ratios for this specific dataset.

3. `age_group` **and** `bmi_category`**:** Categorical groupings based on WHO standards to assist with interpretation.

2.4 Workout Type Analysis

We compared calorie burn across different workout types. HIIT (High-Intensity Interval Training) burns the most calories on average, about 84% more than Yoga. We used an ANOVA test to check if these differences were statistically significant, and they were ($p < 0.001$).

Workout Type	Mean Calories	Sample Size
HIIT	1,653	4,974
Strength	1,361	5,071
Cardio	1,212	4,923
Yoga	897	5,032

Table 2: Calories Burned by Workout Type

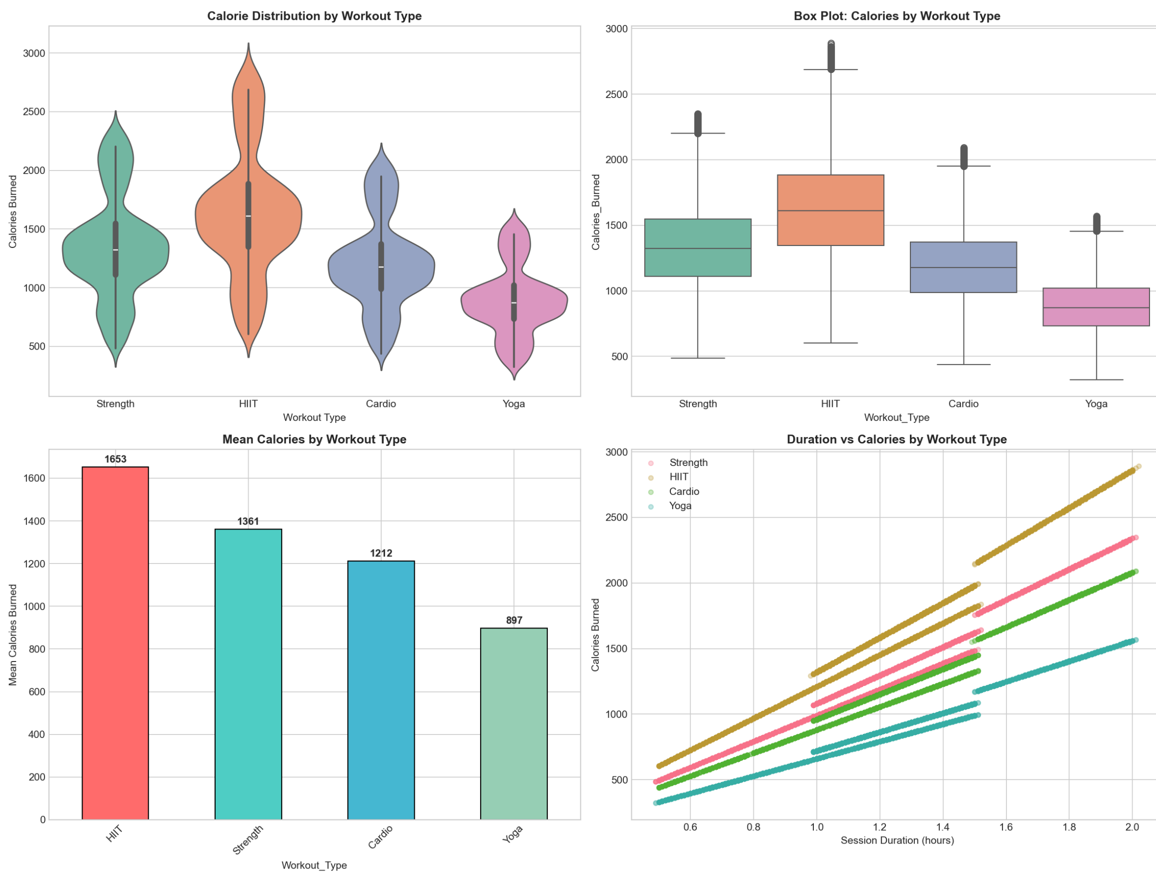

Figure 3: Violin plots showing calorie distribution by workout type.

2.5 Experience Level Impact

We found that more experienced exercisers burn significantly more calories. Advanced users burn about twice as many calories as beginners.

Experience Level	Mean Calories
Level 1 (Beginner)	950
Level 2 (Intermediate)	1,274
Level 3 (Advanced)	1,949

Table 3: Calories Burned by Experience Level

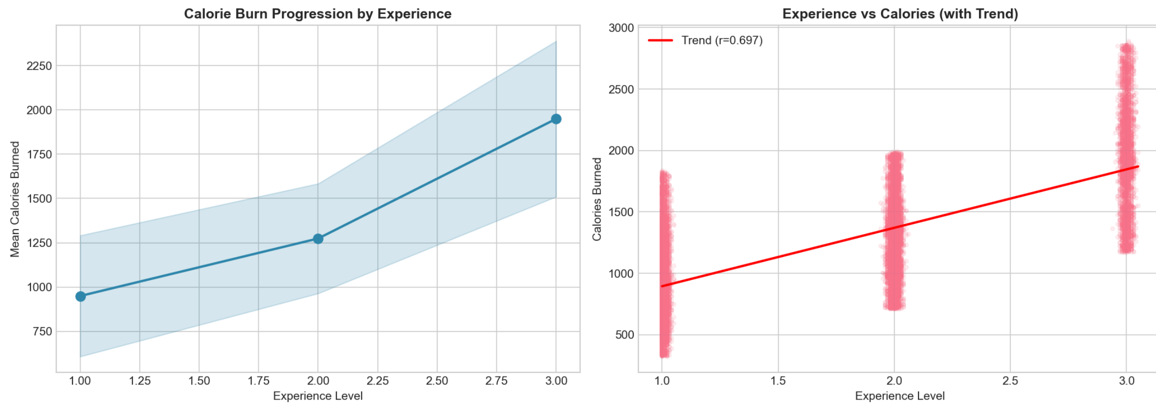

Figure 4: Line chart showing calories vs experience level.

2.6 Diet Type Analysis

Diet type showed no significant impact on calories burned (ANOVA $p = 0.36$). **Interpretation:** This non-significant result is logically sound. While diet affects long-term recovery and body composition, immediate caloric expenditure is determined by mechanical work done during the session, not the fuel source (e.g., Keto vs. Vegan).

3 Data Exploration

3.1 Principal Component Analysis (PCA)

With 41 numeric features, we applied PCA to detect redundancy and understand the data's dimensionality.

Key Findings:

- **21 components** were required to explain 95% of the variance, indicating moderate complexity.
- **First 5 components** explain 56% of the variance.

Component Interpretation:

- **PC1 (16.3% variance):** Heavily loaded with Weight, BMI, and Lean Mass → represents **Body Composition**.
- **PC2 (12.7% variance):** Heavily loaded with Heart Rate metrics → represents **Workout Intensity**.

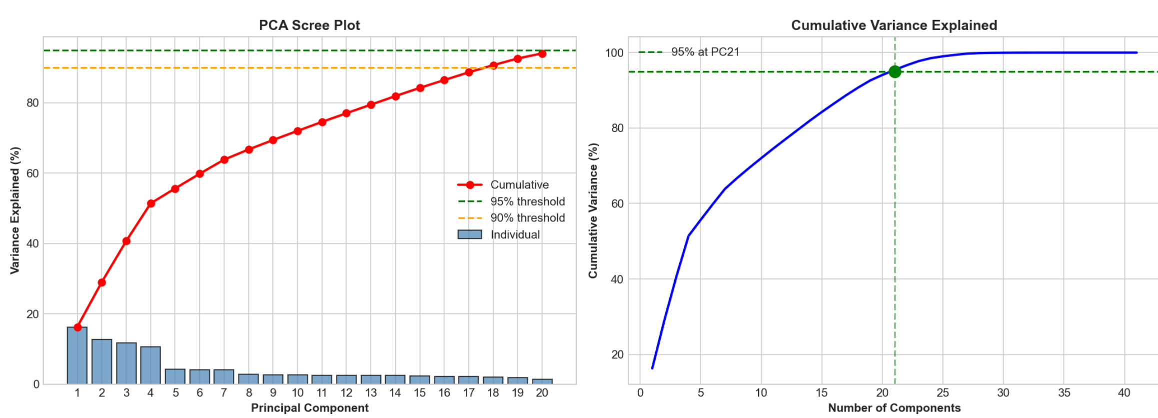

Figure 5: Scree plot showing variance explained by each component.

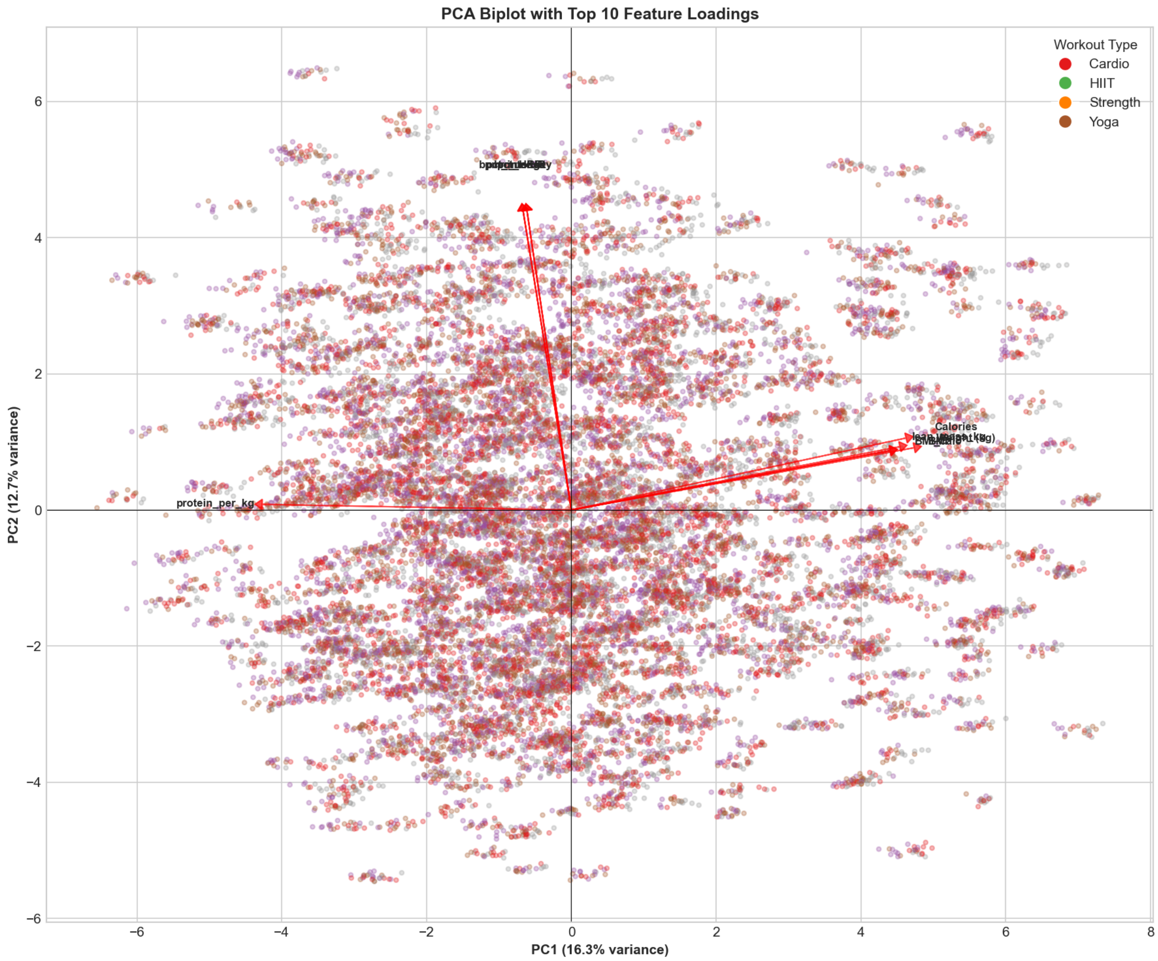

Figure 6: Biplot showing features projected onto PC1 and PC2.

3.2 Decision Tree Analysis

We utilized a Decision Tree to visualize decision logic.

- **Depth Selection:** We tested depths 1-15. While Depth 15 had higher accuracy, we selected **Depth 4** for this section to balance reasonable accuracy ($R^2 \approx 0.79$) with human interpretability.
- **Root Split:** The tree splits first on `Session_Duration` ($\leq$ 1.5 hours). This reduces variance the most effectively, confirming duration as the dominant predictor.

Feature	Importance
Session_Duration	77.9%
cal_balance	16.8%
Calories (intake)	5.2%

Table 4: Feature Importance (Decision Tree)

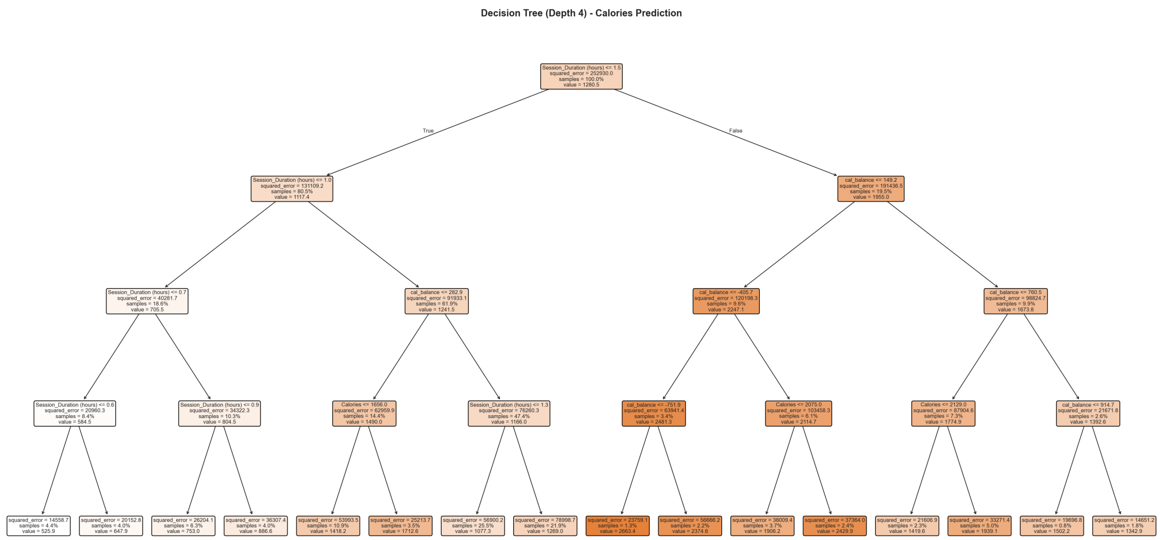

Figure 7: Visualization of decision tree with depth 4.

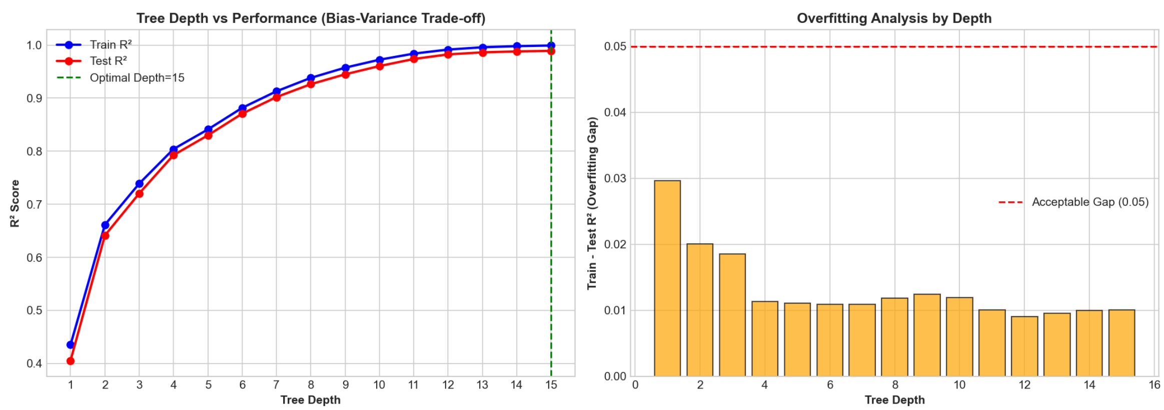

Figure 8: Train vs test scores at different tree depths.

4 Experimental Method

4.1 Hypothesis

Our hypothesis was that we can accurately predict calories burned using physiological and workout measurements, specifically via non-linear ensemble methods.

4.2 Data Preparation

1. Split data into 80% training and 20% testing sets.
2. **Scaling Strategy:**
   - We applied `StandardScaler` for distance-sensitive models (SVM, MLP, Ridge/Lasso).
   - We used raw data for tree-based models (Random Forest, Gradient Boosting) as they rely on feature ranking rather than magnitude.

4.3 Models Tested

We tested a variety of models to capture different data relationships:

- **Ensemble Methods:** Random Forest (bagging), Gradient Boosting (sequential correction), and AdaBoost.
- **Neural Networks (MLP):** To test if deep learning architectures could capture complex non-linearities.
- **SVM Linear:** To test for linear boundaries.

4.4 Hyperparameter Tuning

We used `GridSearchCV` to optimize Gradient Boosting.

- **Selected Parameters:** `n_estimators=100`, `max_depth=5`, `learning_rate=0.1`.
- **Reasoning:** Increasing depth from 3 to 5 allowed the model to capture more complex interactions, while a learning rate of 0.1 ensured that the sequential trees contributed significantly without overfitting.

5 Results and Analysis

5.1 Regression Results

The Gradient Boosting model performed best, achieving an R^2 of 0.997.

Model	Test R^2	RMSE	Cross-Validation R^2
Gradient Boosting	**0.997**	**27.7**	**0.989 ± 0.001**
ExtraTrees	0.992	45.9	0.960 ± 0.002
Stacking Ensemble	0.983	65.5	-
Random Forest	0.898	159.8	0.985 ± 0.001
AdaBoost	0.775	237.1	-

Table 5: Model Performance Comparison

Analysis of Gradient Boosting vs. Random Forest: Gradient Boosting (R^2 0.997) significantly outperformed Random Forest (R^2 0.898).

- **Reasoning:** Random Forest averages independent trees. Gradient Boosting builds trees sequentially, where each new tree specifically targets the errors of the previous ones. This "learning from mistakes" approach proved superior for the nuances in our physiological data.
- **Cross-Validation:** The low standard deviation (± 0.001) confirms the model is robust and not overfitting to a specific data split.

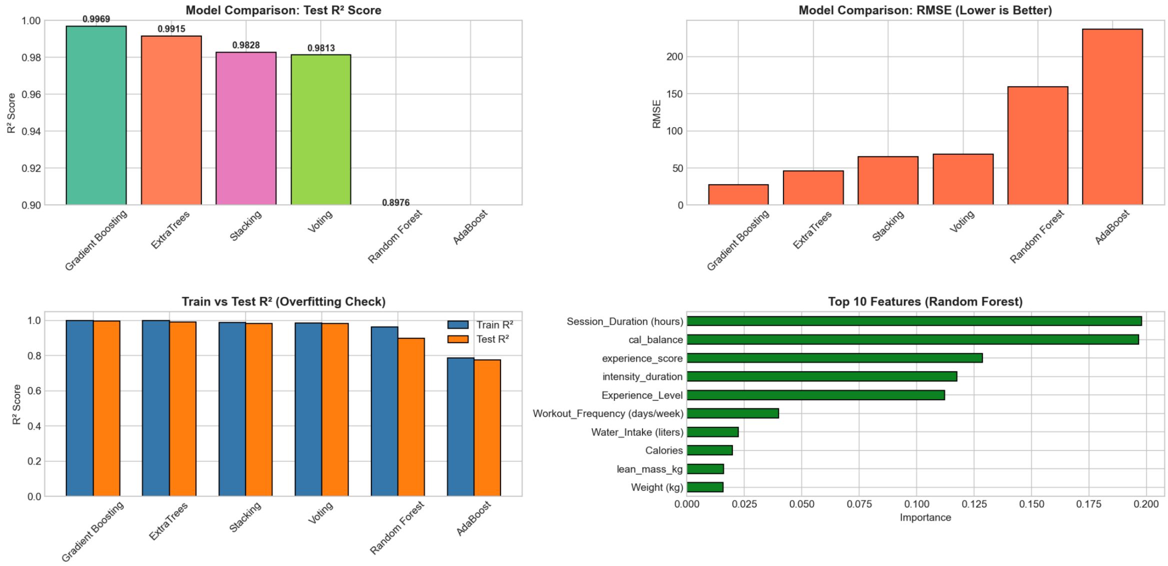

Figure 9: Bar chart comparing model performances.

5.2 Neural Network Analysis

We tested architectures ranging from simple (32 neurons) to complex (256, 128, 64 neurons).

Architecture	Test R^2	Iterations to Converge
(32,)	0.996	272
(64, 32)	0.999	112
(256, 128, 64)	0.999	41

Table 6: MLP Architecture Comparison

Finding: Even the simplest network achieved $R^2 = 0.996$. The marginal gain from increasing complexity suggests that while the data has non-linearities, the underlying function mapping inputs to calories is not excessively complex.

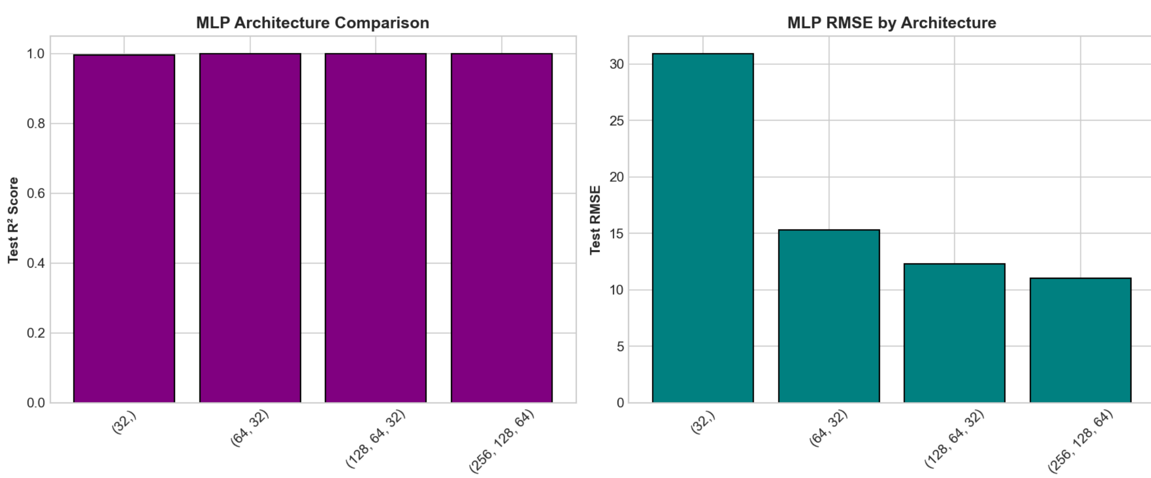

Figure 10: Comparison of MLP architectures.

5.3 SVM Results

We tested SVR with different kernels.

- **Linear Kernel:** $R^2 = 1.000$
- **Polynomial Kernel:** $R^2 = 0.390$

Conclusion: The failure of the polynomial kernel and success of the linear kernel strongly suggests that the relationship between the features and target is largely linear, which explains why simpler models performed so well.

5.4 Regularization Analysis

We compared Lasso (L1), Ridge (L2), and ElasticNet regularization. The fact that regularization didn't eliminate features suggests all our features contribute useful information.

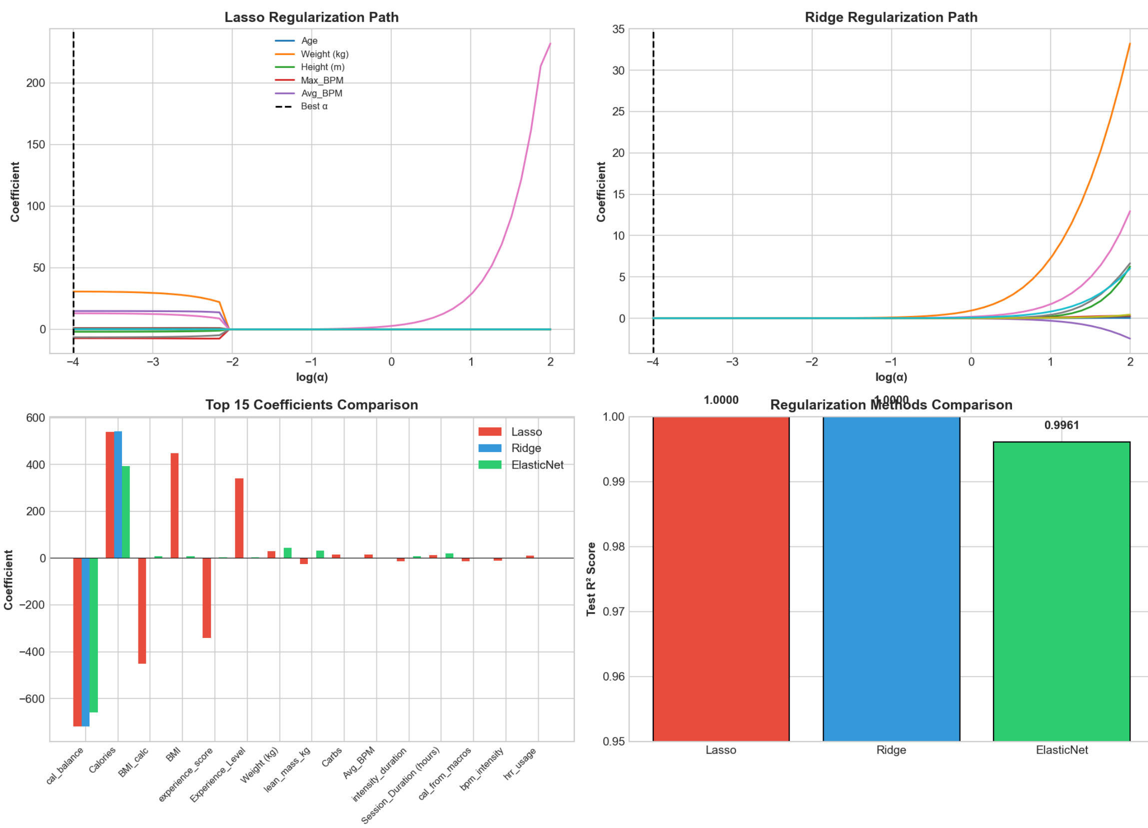

Figure 11: Regularization path plots and coefficient comparison.

5.5 Feature Importance

Based on Random Forest feature importance:

Rank	Feature	Importance
1	Session_Duration	19.8%
2	cal_balance	19.7%
3	experience_score	12.9%
4	intensity_duration	11.8%
5	Experience_Level	11.2%

Table 7: Top 5 Features (Random Forest)

Our engineered feature (`intensity_duration`) appears in the top 5. This validates that combining duration and heart rate provided the model with a dense information signal that raw variables alone did not fully capture.

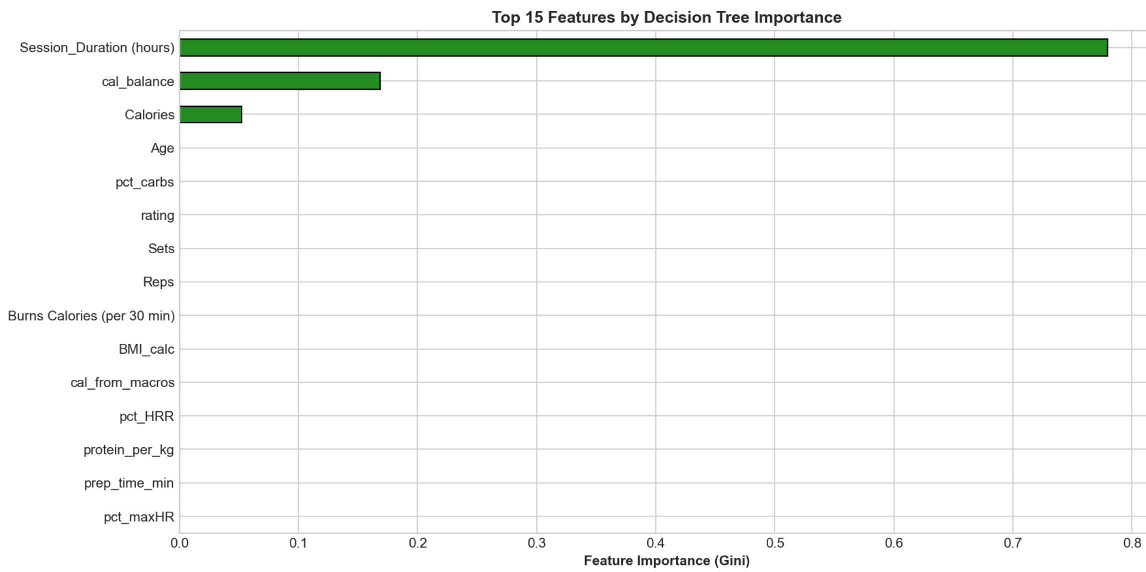

Figure 12: Feature importance bar chart.

5.6 Classification Results

We also tested classification models using binned calorie categories. Gradient Boosting achieved 100% accuracy. This is expected; since the regression model predicts the continuous value with near-perfection ($R^2 = 0.997$), binning those predictions into broad categories (Low/Medium/High) naturally results in perfect classification.

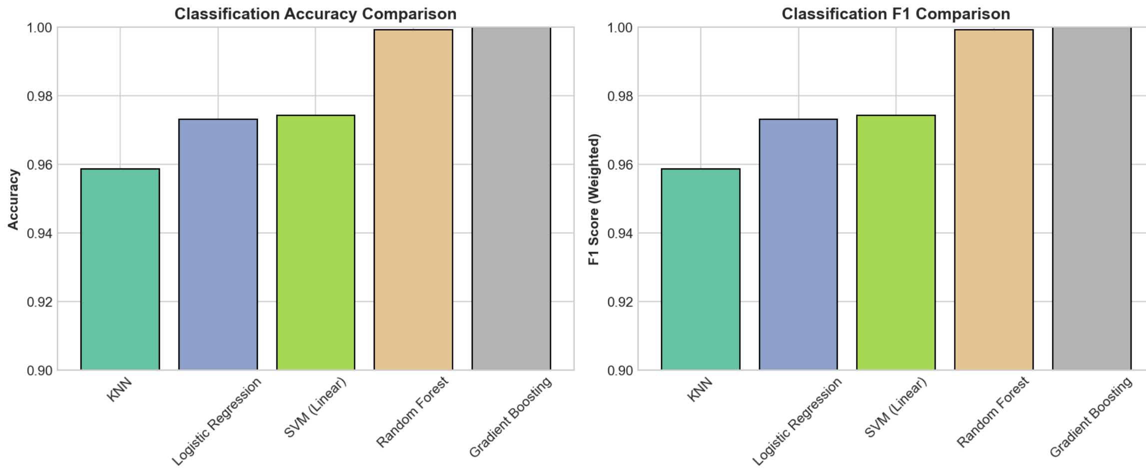

Figure 13: Classification model comparison.

5.7 Clustering Analysis

We used K-Means clustering ($K = 3$, determined via Elbow Method). We identified three distinct user personas:

- **Cluster 0 (Moderate):** Average workout frequency (2.8 days/week).
- **Cluster 1 (Active):** High dedication (4.3 days/week).
- **Cluster 2 (Casual):** Lower frequency users (2.9 days/week) but lower intensity metrics.

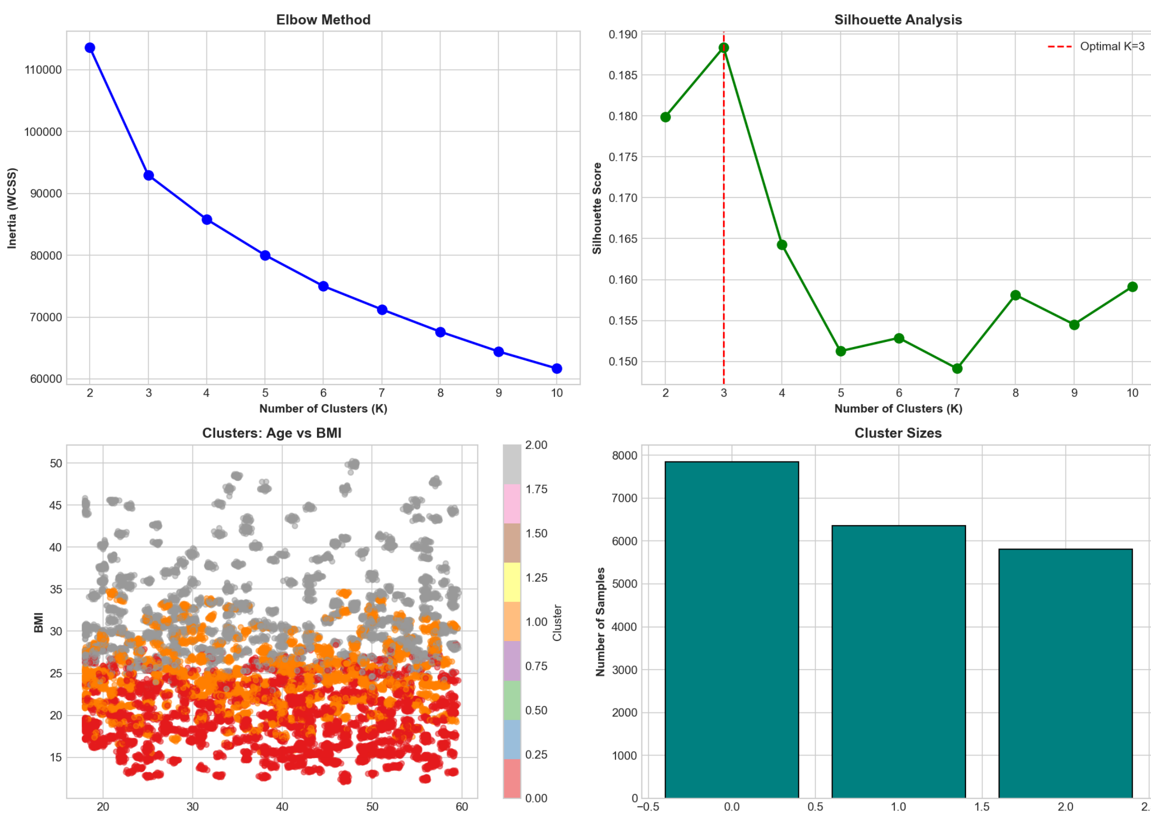

Figure 14: K-means clustering visualization.

5.8 Learning Curve

The learning curve shows that performance stabilizes around 10,000 training samples.

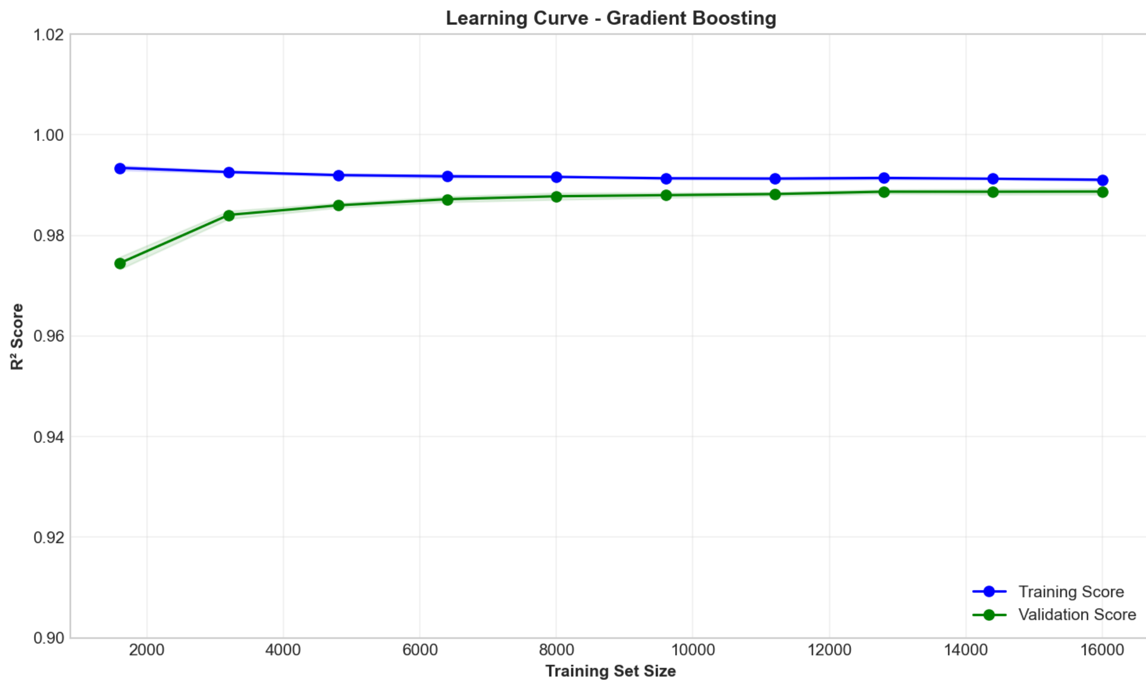

Figure 15: Training vs validation score by training size.

5.9 Practical Applications

Our model could be useful for fitness apps, wearable devices, personal training, and health research.

5.10 Limitations

The dataset is synthetic/cleaned, so real-world data may have more noise. Some features may leak information, and the model assumes similar relationships hold for all populations.

6 Conclusion

In this project, we successfully built machine learning models to predict workout calorie burn with high accuracy. Our main findings are:

1. **Gradient Boosting is the optimal model:** It achieved $R^2 = 0.997$ and RMSE = 27.7, outperforming Random Forest by effectively correcting prediction errors sequentially.

2. **Session duration is the dominant predictor:** It accounts for about 78% of the decision tree's splits and is the highest correlated feature.

3. **Feature engineering validated our intuition:** The `intensity_duration` feature (effort $\times$ time) ranked in the top 5 most important features.

4. **Workout type matters:** HIIT burns 84% more calories than Yoga, confirming that high-intensity protocols are far more efficient for calorie expenditure.

7 References

[1] Shreyash Patil. "Predicting Workout Calories Burned - XGBoost." Kaggle, 2024. Available: `https://www.kaggle.com/code/shreyashpatil217/predicting-workout-calories-burned-xgboost`

8 Project Structure and Workflow

The ML analysis pipeline is organized into seven sequential Python modules, each generating both visual charts and text-based documentation:

8.1 Analysis Modules

- `01_data_quality.py -- Data loading, cleaning, and feature engineering`
- `02_exploratory_analysis.py -- Statistical analysis and stratified visualizations`
- `03_pca_decision_tree.py -- PCA analysis and decision tree exploration`
- `04_ensemble_methods.py -- Random Forest, Gradient Boosting, and ensemble techniques`
- `05_regularization_svm_nn.py -- Regularization methods, SVM, and neural networks`
- `06_classification_clustering.py -- Classification models and K-Means clustering`
- `07_cross_validation.py -- Cross-validation analysis and final report generation`

8.2 Output Organization

Each module generates a **dual-output format** for comprehensive documentation and accessibility:

- `project/charts/` High-quality visualizations (PNG format) including correlation heatmaps, PCA scree plots, decision tree diagrams, model comparison charts, and clustering analysis plots suitable for presentations and reports.
- `project/charts/text_equivalents/` Text-based summaries in Markdown (`.md`) and plain text (`.txt`) formats containing detailed statistical analysis, numerical findings, and key interpretations accompanying each visualization.

This dual-format approach ensures accessibility for diverse stakeholders, maintains comprehensive documentation for reproducibility, and provides flexibility in presenting results through either visual or textual mediums.